

Important: Text generated by AI and may contain incorrect information, always check the content.

```
```python
Import necessary libraries
import matplotlib.pyplot as plt
import seaborn as sns

Check for null or categorical values
display(df.isnull().sum())
display(df.dtypes)

Since there are no null values and all columns are numerical, we can proceed with visualizations

Histograms to understand the distribution of individual features
display('Histograms for individual features:')
fig, axes = plt.subplots(nrows=11, ncols=2, figsize=(20, 40))
for i, column in enumerate(df.columns):
 ax = axes[i // 2, i % 2]
 sns.histplot(df[column], ax=ax, kde=True)
 ax.set_title(column)
plt.tight_layout()
display(fig)

Boxplots to understand the distribution and outliers
display('Boxplots for individual features:')
fig, axes = plt.subplots(nrows=11, ncols=2, figsize=(20, 40))
for i, column in enumerate(df.columns):
 ax = axes[i // 2, i % 2]
 sns.boxplot(df[column], ax=ax)
 ax.set_title(column)
plt.tight_layout()
display(fig)

Correlation heatmap to understand the relationship between features
display('Correlation heatmap:')
plt.figure(figsize=(20, 18))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', square=True)
plt.title('Correlation Heatmap')
display(plt.gcf())

Pairplot to understand the relationship between features
Since there are 22 features, a pairplot with all features may not be very informative
```



```
We will consider a sample of features for the pairplot
sample_features = ['baseline value', 'fetal_health', 'histogram_mean', 'histogram_median', 'abnormal_short_term_variability']
display('Pairplot for selected features:')
sns.pairplot(df[sample_features])
display(plt.gcf())
````
```

| | |
|--|---|
| | 0 |
| baseline value | 0 |
| accelerations | 0 |
| fetal_movement | 0 |
| uterine_contractions | 0 |
| light_decelerations | 0 |
| severe_decelerations | 0 |
| prolongued_decelerations | 0 |
| abnormal_short_term_variability | 0 |
| mean_value_of_short_term_variability | 0 |
| percentage_of_time_with_abnormal_long_term_variability | 0 |
| mean_value_of_long_term_variability | 0 |
| histogram_width | 0 |
| histogram_min | 0 |
| histogram_max | 0 |
| histogram_number_of_peaks | 0 |
| histogram_number_of_zeroes | 0 |
| histogram_mode | 0 |





0

| | |
|--------------------|---|
| histogram_mean | 0 |
| histogram_median | 0 |
| histogram_variance | 0 |
| histogram_tendency | 0 |
| fetal_health | 0 |

dtype: int64

0

| | |
|--|---------|
| baseline value | float64 |
| accelerations | float64 |
| fetal_movement | float64 |
| uterine_contractions | float64 |
| light_decelerations | float64 |
| severe_decelerations | float64 |
| prolongued_decelerations | float64 |
| abnormal_short_term_variability | float64 |
| mean_value_of_short_term_variability | float64 |
| percentage_of_time_with_abnormal_long_term_variability | float64 |
| mean_value_of_long_term_variability | float64 |
| histogram_width | float64 |
| histogram_min | float64 |



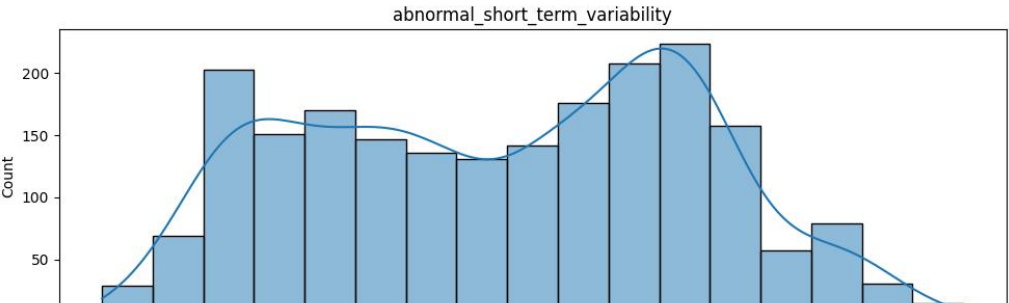
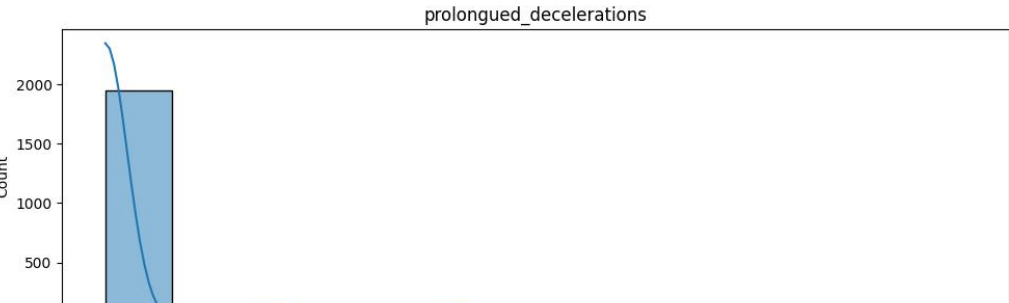
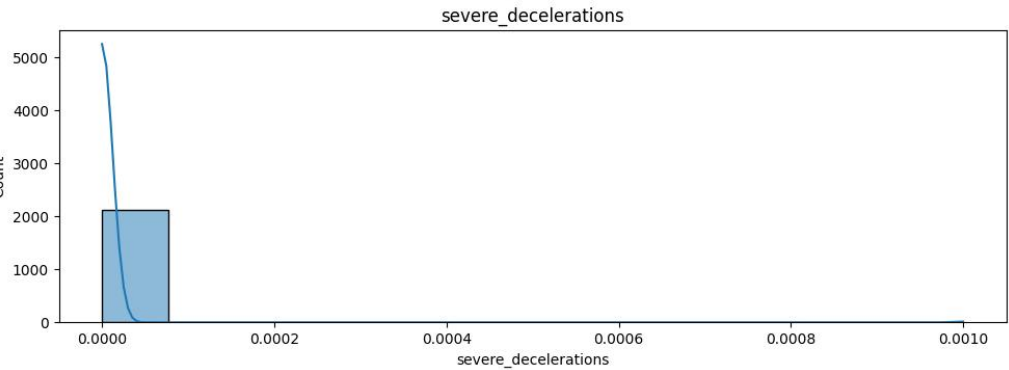
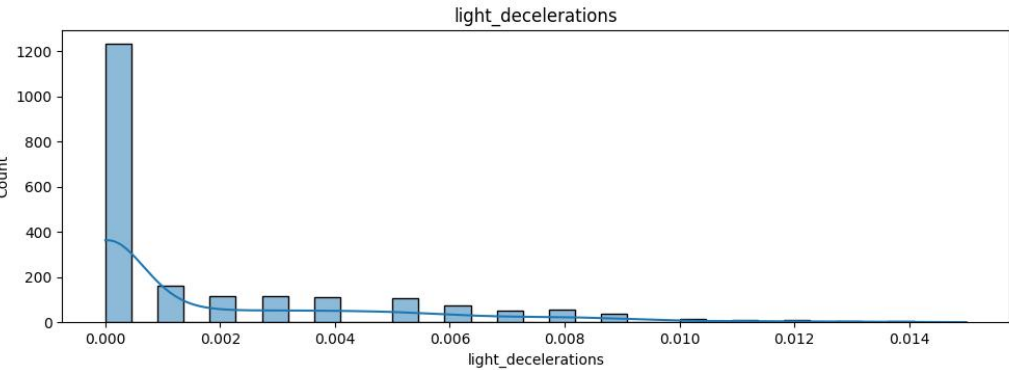
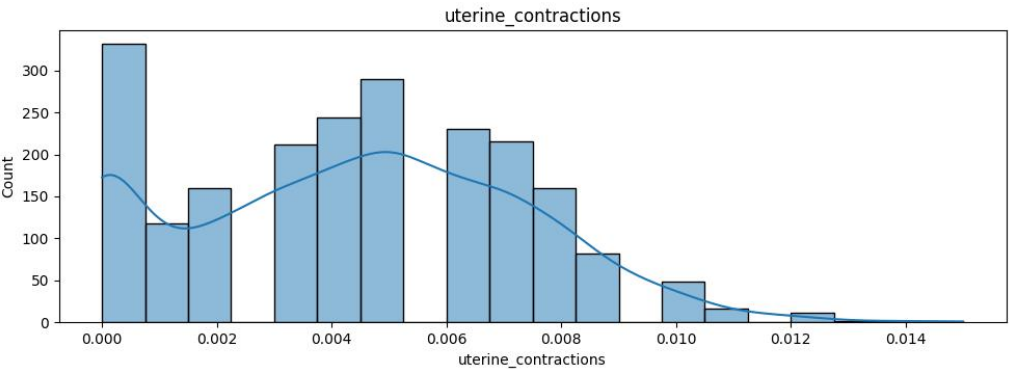
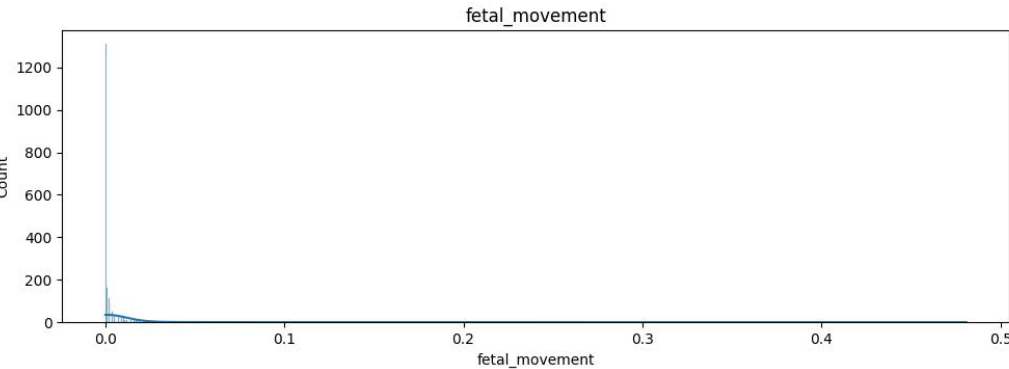
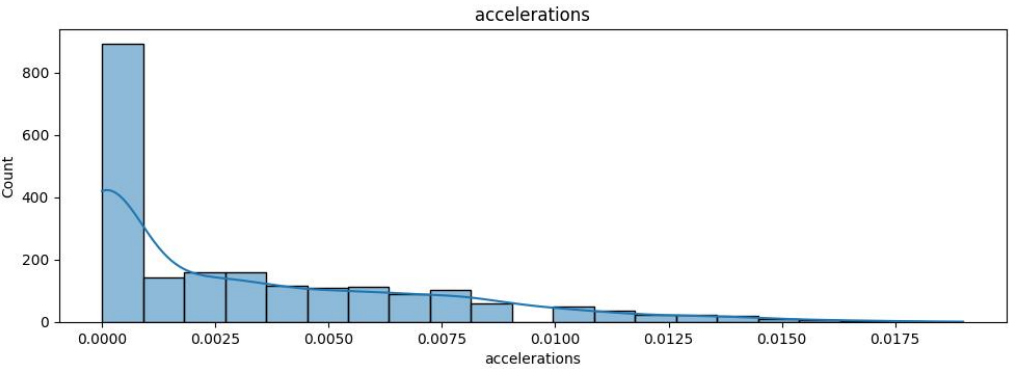
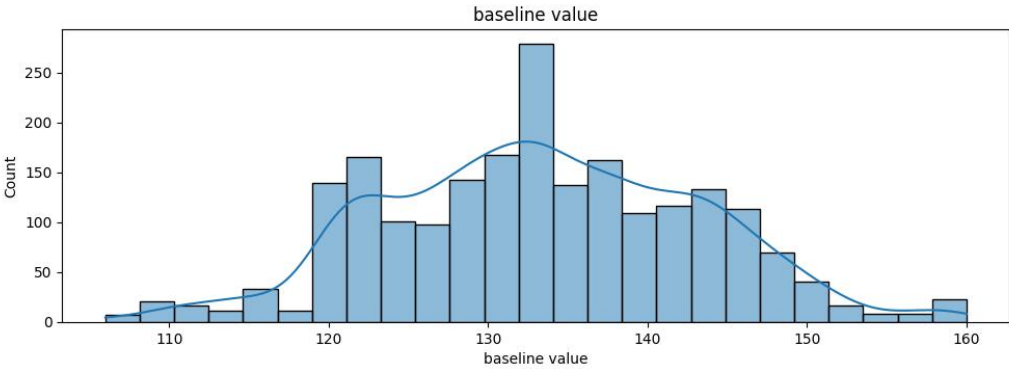


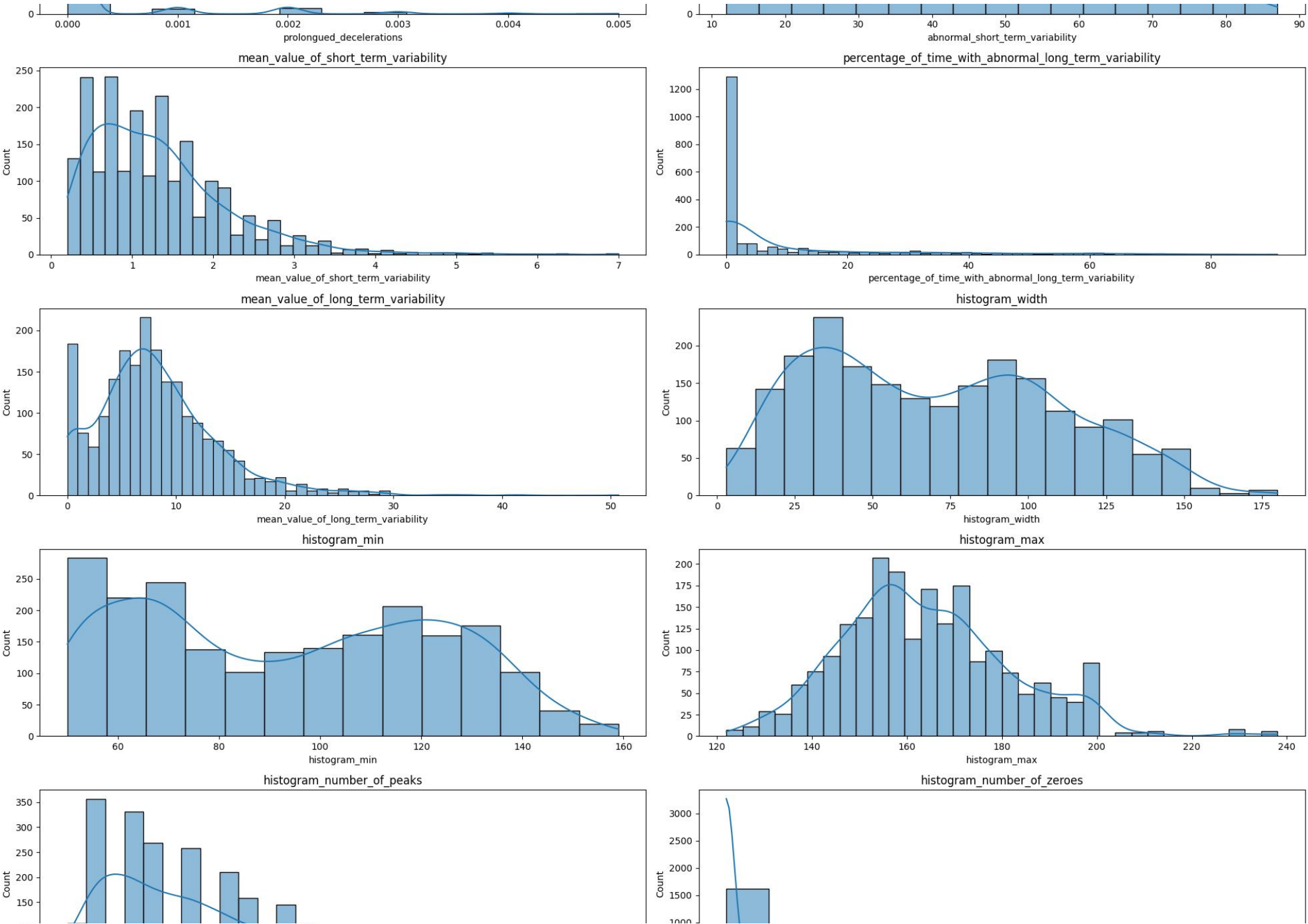
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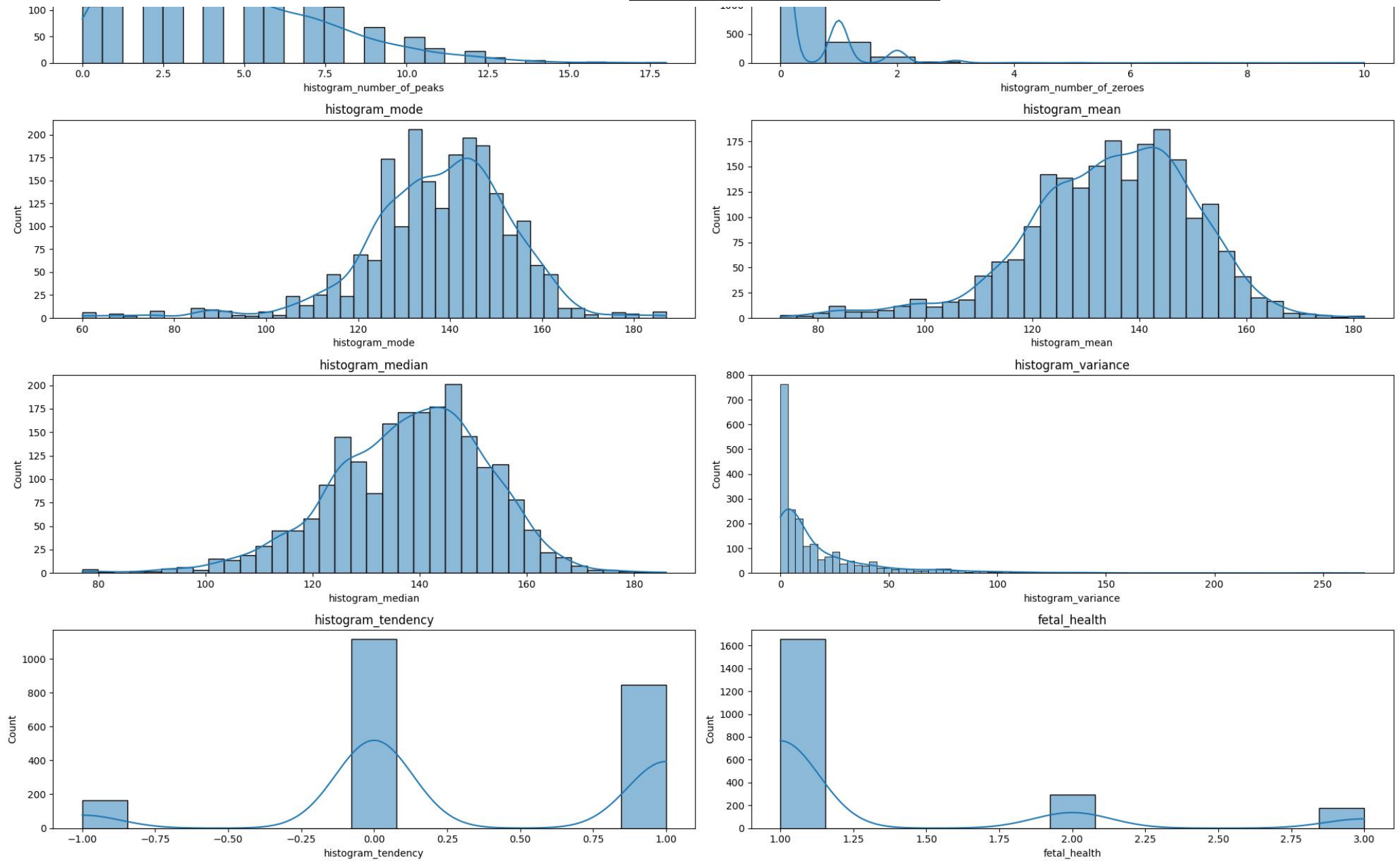
| | |
|----------------------------|---------|
| histogram_max | float64 |
| histogram_number_of_peaks | float64 |
| histogram_number_of_zeroes | float64 |
| histogram_mode | float64 |
| histogram_mean | float64 |
| histogram_median | float64 |
| histogram_variance | float64 |
| histogram_tendency | float64 |
| fetal_health | float64 |

dtype: object
'Histograms for individual features:'

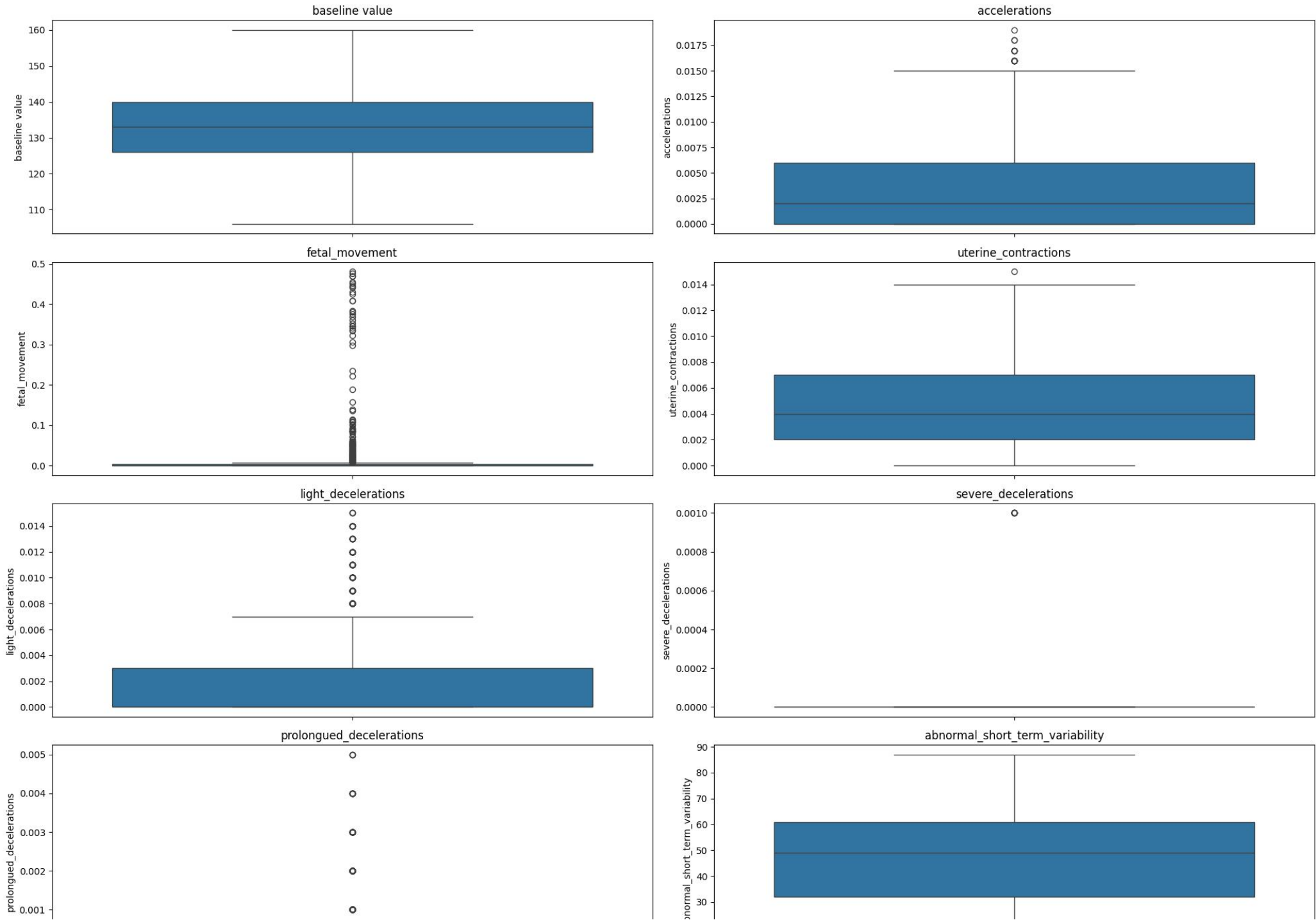
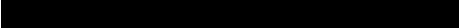


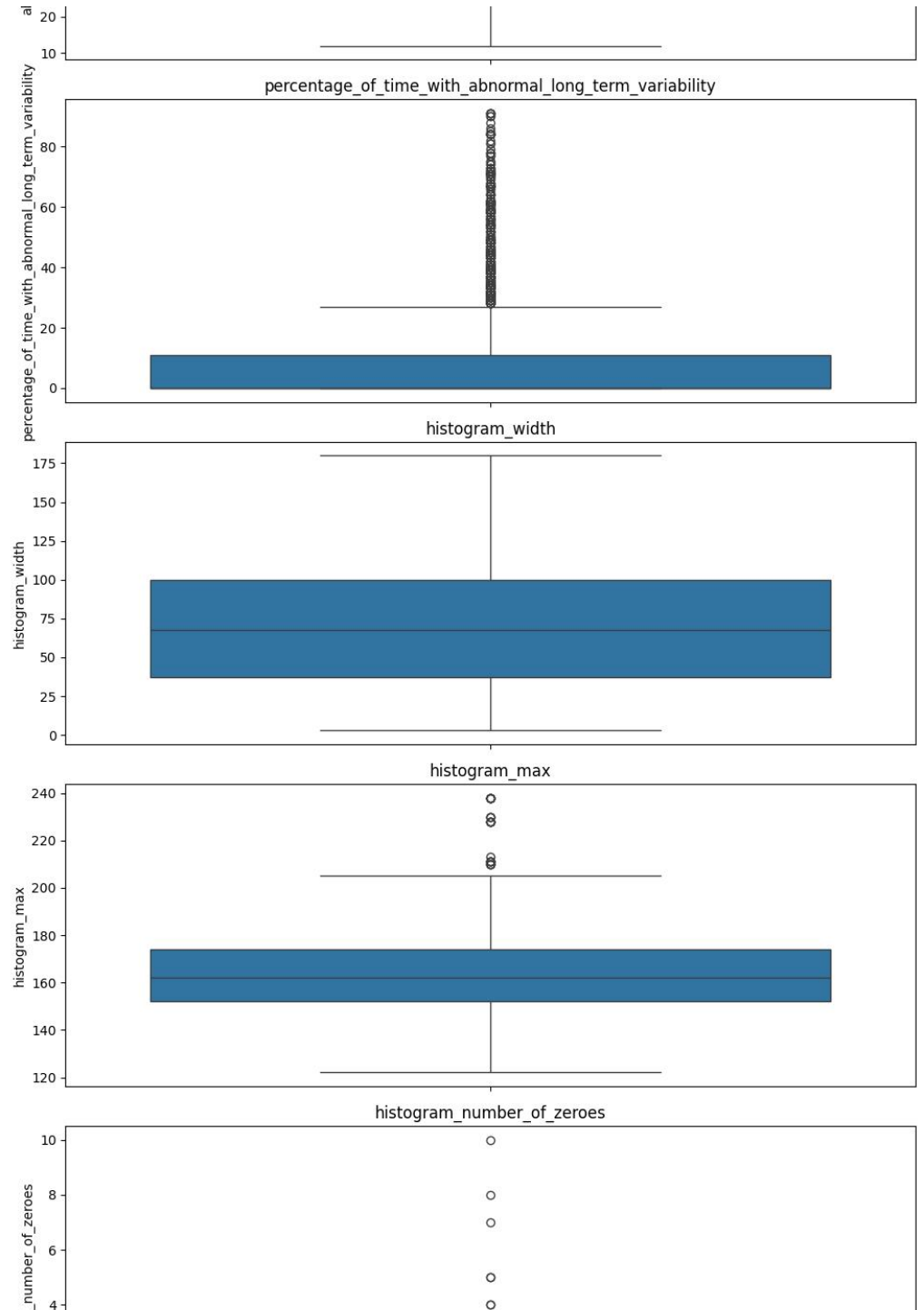
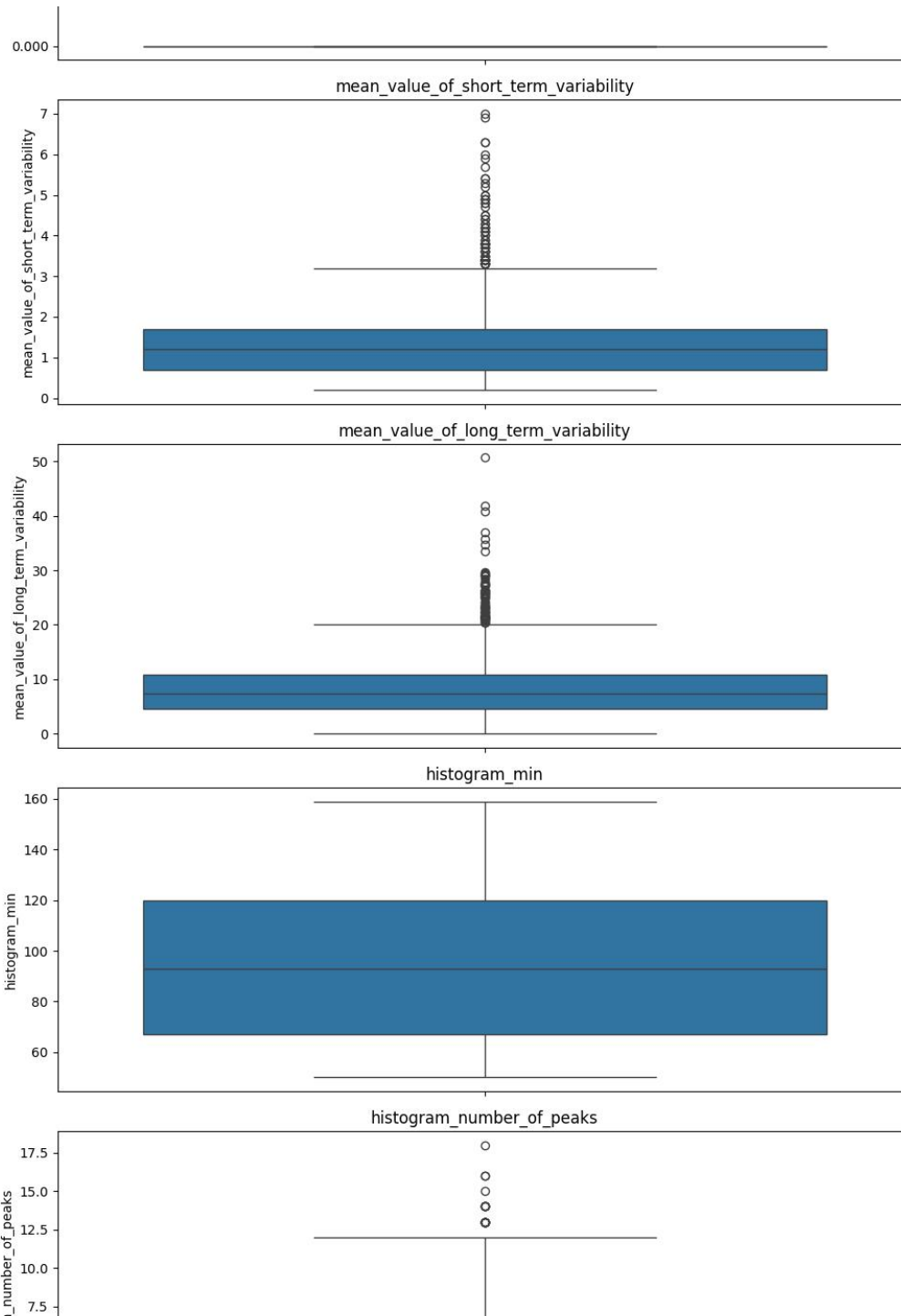


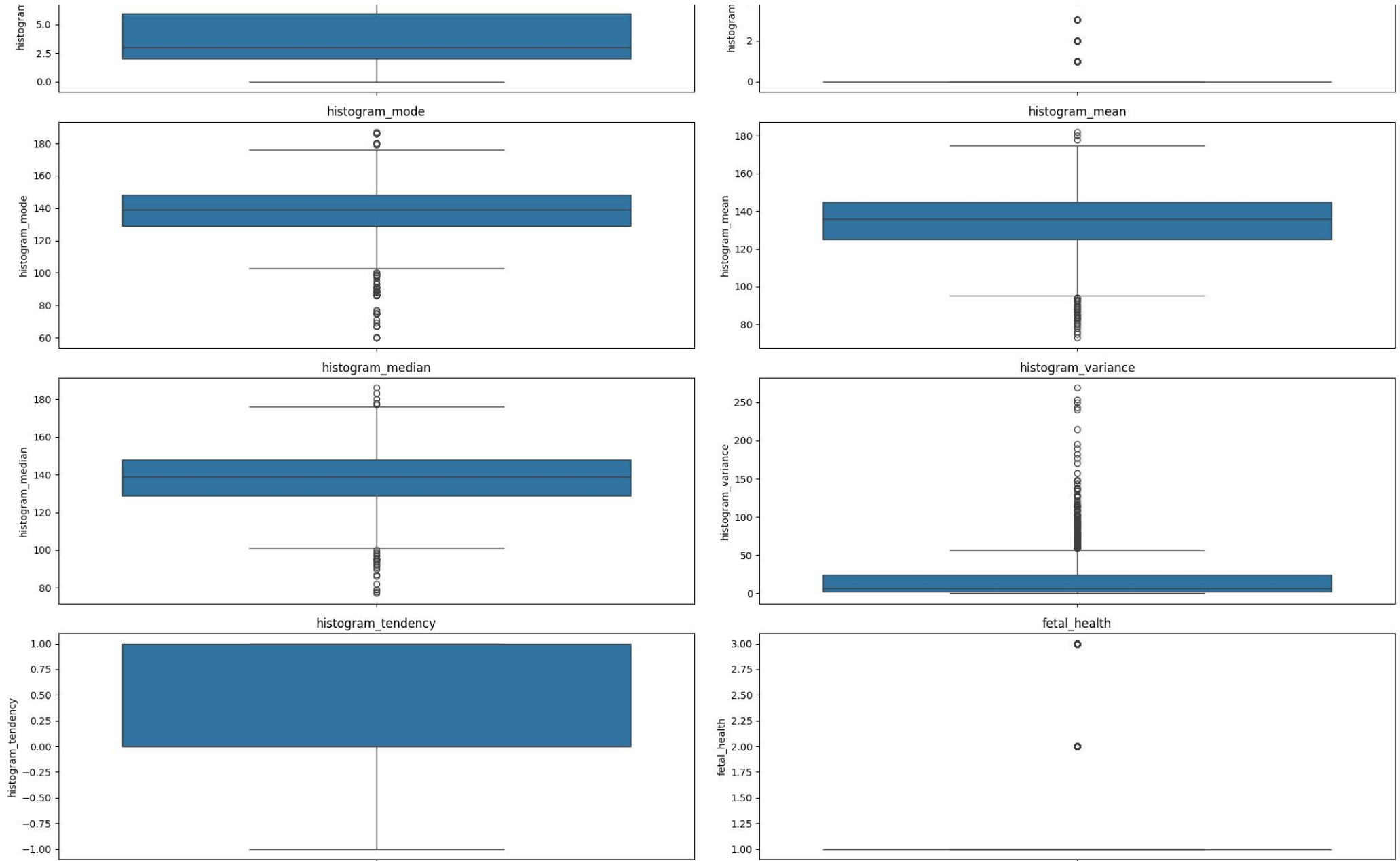




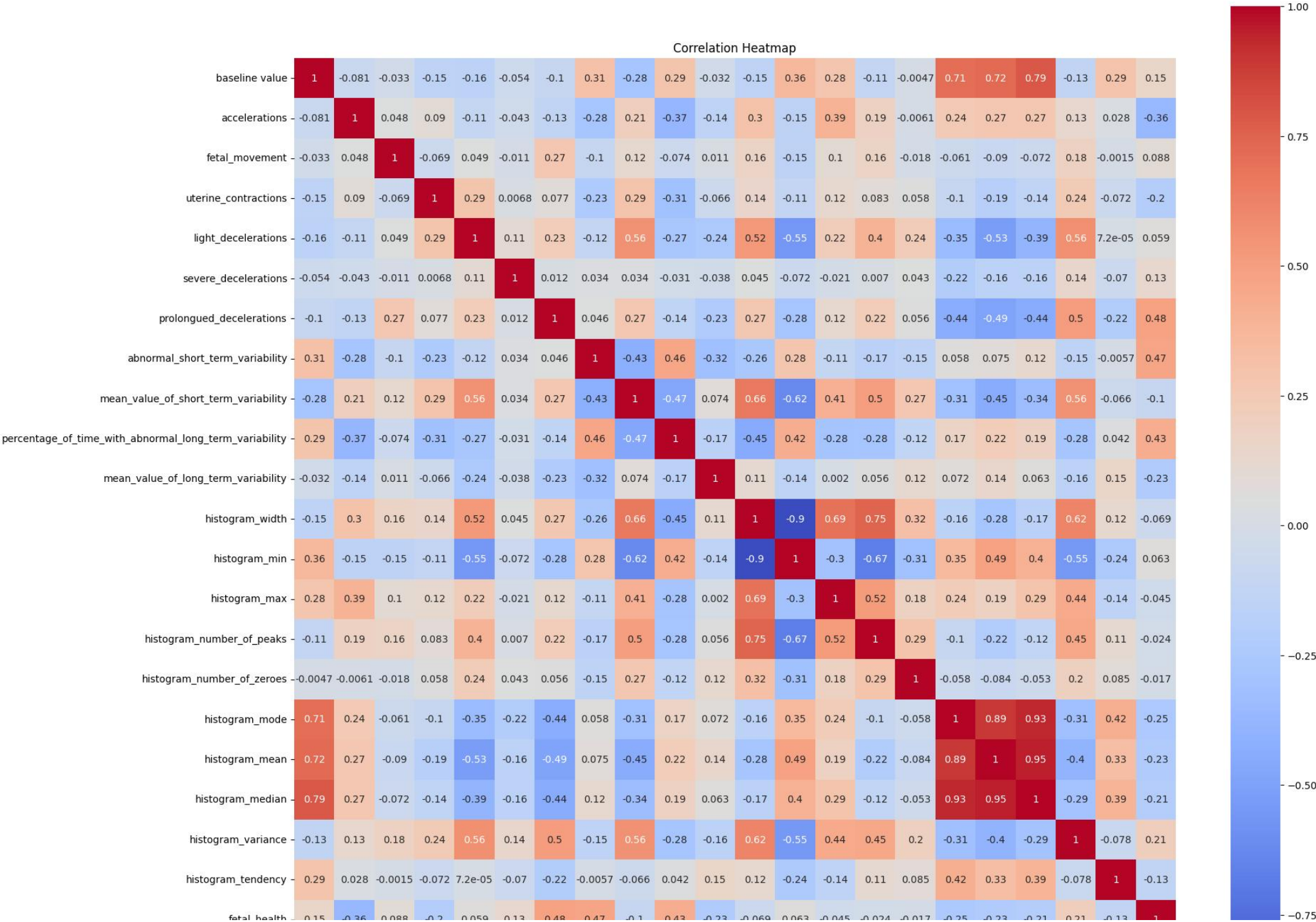
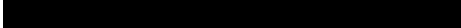
'Boxplots for individual features:'







'Correlation heatmap:'



'Pairplot for selected features:'

